

```
!pip install ultralytics
```

```
from ultralytics import YOLO
from IPython.display import Image, display
import os
```

Collecting ultralytics

Downloading ultralytics-8.4.66-py3-none-any.whl.metadata (41 kB)

41.3/41.3 kB 2.8 MB/s eta 0:00:00

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
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 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1

```
print(os.listdir())
```

```
['.config', 'apple_test.jpg', 'banana_test.jpeg', 'orange_test.jpg', 'potato_test.jpeg', 'best (1).pt', 'sample_data']
```

```
model = YOLO("best (1).pt")
```

```
print("Model Loaded Successfully")
```

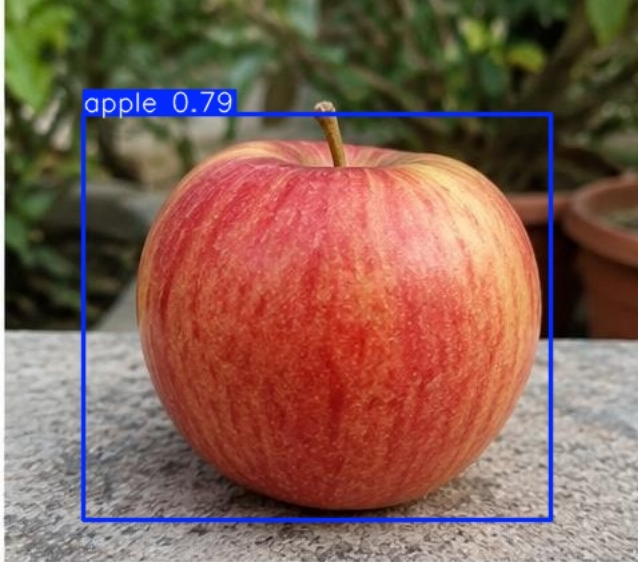
```
Model Loaded Successfully
```

```
result = model.predict(
    source="apple_test.jpg",
    save=True,
    conf=0.25,
    project="final_testing",
    name="apple",
    exist_ok=True
)
```

```
image 1/1 /content/apple_test.jpg: 576x640 1 apple, 95.9ms  
Speed: 15.1ms preprocess, 95.9ms inference, 51.1ms postprocess per image at shape (1, 3, 576, 640)  
Results saved to /content/runs/detect/final_testing/apple
```

```
import os  
  
print(os.listdir("runs/detect/final_testing/apple"))  
  
['apple_test.jpg']
```

```
from IPython.display import Image, display  
  
display(Image(filename="runs/detect/final_testing/apple/apple_test.jpg"))
```



```
result = model.predict(  
    source="banana_test.jpeg",  
    save=True,  
    conf=0.25,  
    project="runs/detect/final_testing",  
    name="banana",  
    exist_ok=True  
)
```

```
image 1/1 /content/banana_test.jpeg: 640x544 1 banana, 48.0ms  
Speed: 3.3ms preprocess, 48.0ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 544)  
Results saved to /content/runs/detect/runs/detect/final_testing/banana
```

```
import os  
  
print(os.listdir("runs/detect/runs/detect/final_testing/banana"))  
  
['banana_test.jpg']
```

```
from IPython.display import Image, display  
  
display(Image(filename="runs/detect/runs/detect/final_testing/banana/banana_test.jpg"))
```



```
result = model.predict(  
    source="orange_test.jpg",  
    save=True,  
    conf=0.25,  
    project="final_testing",  
    name="orange",  
    exist_ok=True  
)
```

image 1/1 /content/orange_test.jpg: 448x640 1 apple, 1 orange, 85.3ms
Speed: 2.5ms preprocess, 85.3ms inference, 0.6ms postprocess per image at shape (1, 3, 448, 640)
Results saved to /content/runs/detect/final_testing/orange

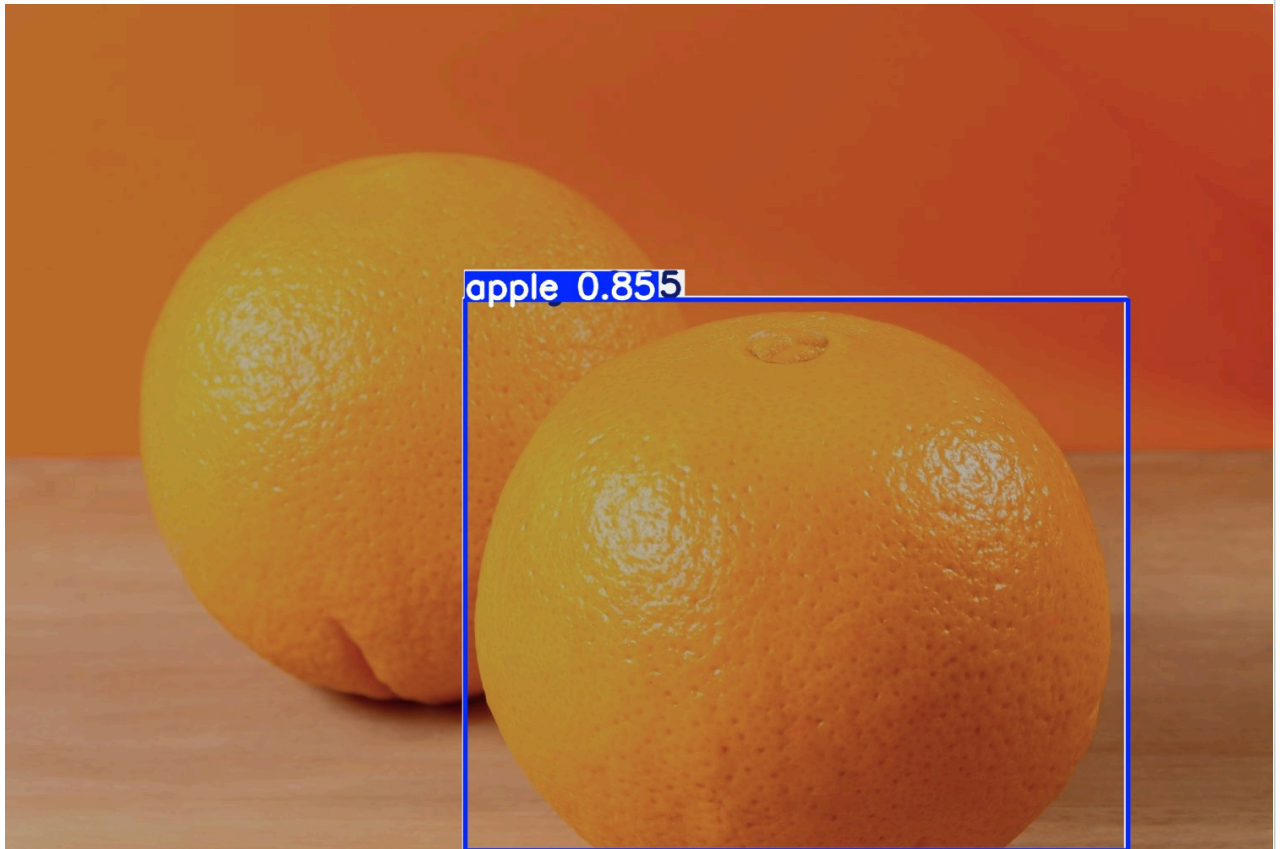

```
import os

print(os.listdir("runs/detect/final_testing/orange"))

['orange_test.jpg']
```

```
from IPython.display import Image, display

display(Image(filename="runs/detect/final_testing/orange/orange_test.jpg"))
```



```
result = model.predict(
    source="potato_test.jpeg",
    save=True,
    conf=0.25,
    project="final_testing",
    name="potato",
    exist_ok=True
)
```

image 1/1 /content/potato_test.jpeg: 640x384 1 potato, 51.2ms
Speed: 3.3ms preprocess, 51.2ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 384)
Results saved to /content/runs/detect/final_testing/potato

```
import os

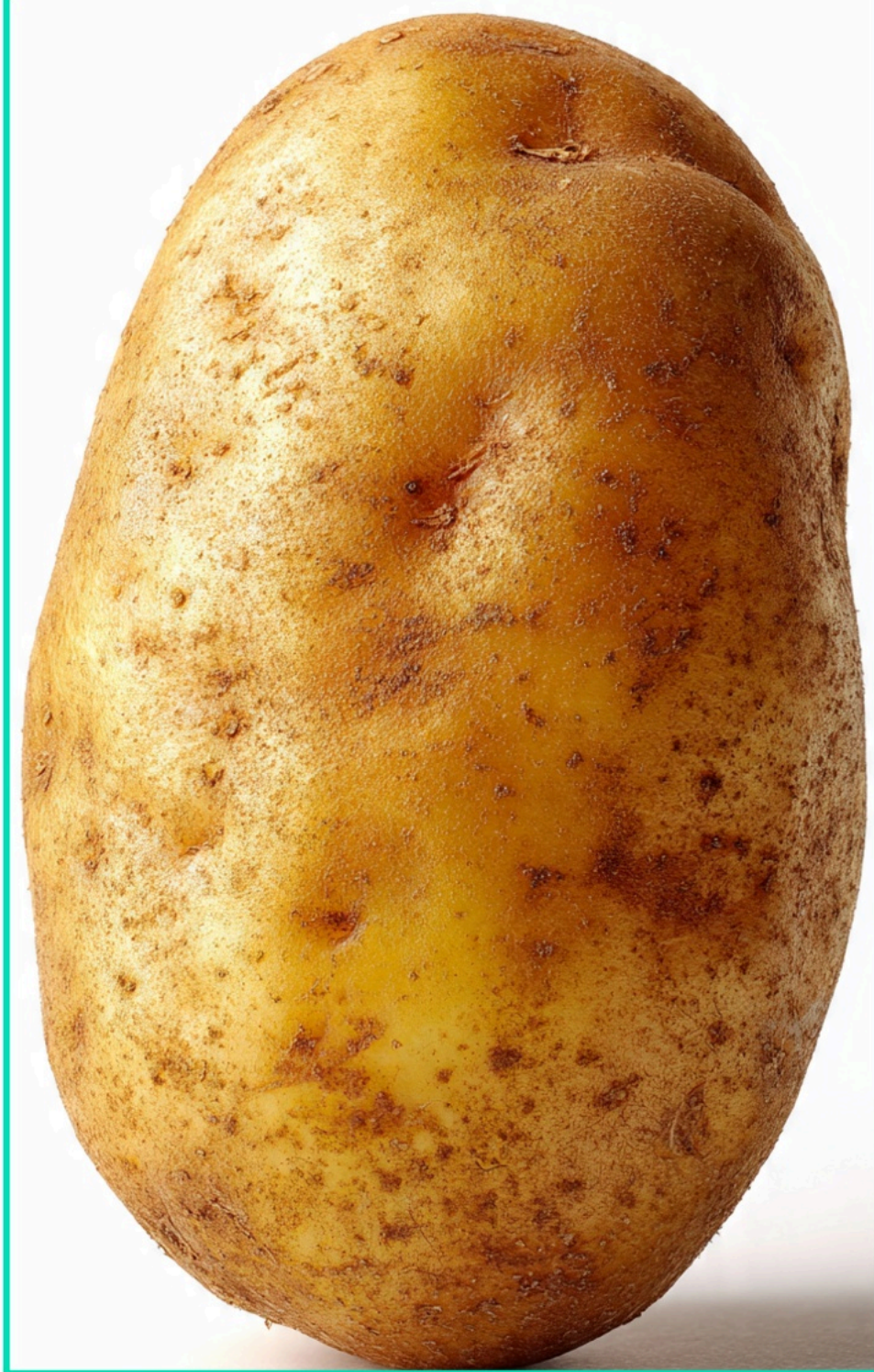
print(os.listdir("runs/detect/final_testing/potato"))

['potato_test.jpg']
```

```
from IPython.display import Image, display

display(Image(filename="runs/detect/final_testing/potato/potato_test.jpg"))
```

potato 0.96



```
!zip -r final_detection_results.zip runs/detect/final_testing
```

```
adding: runs/detect/final_testing/ (stored 0%)
```

```
adding: runs/detect/final_testing/orange/ (stored 0%)
```

```
adding: runs/detect/final_testing/orange/orange_test.jpg (deflated 4%)
```

```
adding: runs/detect/final_testing/potato/ (stored 0%)
adding: runs/detect/final_testing/potato/potato_test.jpg (deflated 3%)
adding: runs/detect/final_testing/apple/ (stored 0%)
adding: runs/detect/final_testing/apple/apple_test.jpg (deflated 3%)
```

```
from google.colab import files

files.download("final_detection_results.zip")
```

```
!find runs -type f

runs/detect/runs/detect/final_testing/banana/banana_test.jpg
runs/detect/final_testing/orange/orange_test.jpg
runs/detect/final_testing/potato/potato_test.jpg
runs/detect/final_testing/apple/apple_test.jpg
```

```
!mkdir -p final_submission/apple
!mkdir -p final_submission/banana
!mkdir -p final_submission/orange
!mkdir -p final_submission/potato

!cp runs/detect/final_testing/apple/apple_test.jpg final_submission/apple/
!cp runs/detect/runs/detect/final_testing/banana/banana_test.jpg final_submission/banana/
!cp runs/detect/final_testing/orange/orange_test.jpg final_submission/orange/
!cp runs/detect/final_testing/potato/potato_test.jpg final_submission/potato/
```

```
!find final_submission -type f

final_submission/orange/orange_test.jpg
final_submission/potato/potato_test.jpg
final_submission/banana/banana_test.jpg
final_submission/apple/apple_test.jpg
```

```
!zip -r final_detection_results.zip final_submission
```

```
adding: final_submission/ (stored 0%)
adding: final_submission/orange/ (stored 0%)
adding: final_submission/orange/orange_test.jpg (deflated 4%)
adding: final_submission/potato/ (stored 0%)
adding: final_submission/potato/potato_test.jpg (deflated 3%)
adding: final_submission/banana/ (stored 0%)
adding: final_submission/banana/banana_test.jpg (deflated 3%)
adding: final_submission/apple/ (stored 0%)
adding: final_submission/apple/apple_test.jpg (deflated 3%)
```